



Robustness assessment of Hetero-functional graph theory based model of interdependent urban utility networks

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ABSTRACT

Increasing urban population imposes a substantial and growing burden on the supporting infrastructure, such as electricity, water, heating, natural gas, road transportation. This paper presents a Hetero-functional graph theory (HFGT) based modeling framework for these integrated infrastructures followed by an analysis of network robustness. The supporting infrastructures along with the infrastructure repair facilities are considered. In contrast to conventional graph representations, a weighted HFGT model is used to capture the system processes and mutual dependencies among resources. To assess robustness of interdependent networks, impacts of complete/partial and random/targeted attacks are quantified. Specifically, various attack scenarios are simulated and the vulnerability of the network is evaluated. Additionally, several robustness metrics are used to provide a comprehensive evaluation of system robustness. The proposed weighted HFGT modeling and robustness assessment approach is tested using a synthetic interdependent network, comprising of an electrical power system, a water network, a district heating network, a natural gas system and a road transportation network. Results demonstrate that system robustness can be enhanced via securing system information and mitigating attack strength.

1. Introduction

1.1. Motivation and background

The world population has increased by 113.3% [1] in the past 50 years. More importantly, with massive urbanization, the percentage of global population in urban areas increased from 34% to 52% in the past five decades [2]. The swelling urban population imposes increasingly intense demands over a variety of infrastructures. A large-scale urban area is comprised of multiple utility networks that deliver a variety of services to the community such as electricity [3], water [4], heating [5], natural gas [6], road transportation [7], and so on. These networks are inherently integrated and their interdependencies are becoming increasingly tight due to the increased reliance on cyber infrastructure to enable smart and efficient operation.

The concepts of resilience and robustness have attracted tremendous attention in recent years in response to the frequent and widespread natural disasters and man-made malicious attacks across the globe. These extreme events impose significant threats to our urban infrastructures [8–10]. Several approaches have been proposed to model and evaluate the robustness of interdependent urban infrastructures such as hierarchical graph representation [11], dynamic network flow model [12], agent based modeling [13], among others. This paper, for

the first time, proposes adopting a “weighted hetero-functional graph theory” concept [14] to model and conduct robustness analysis of interdependent urban infrastructure networks, comprising of electricity, water, heating, natural gas and road transportation networks (along with infrastructure repair services).

1.2. Literature review

The interdependency among infrastructures calls for coordination of the constituent systems. Some examples of integrated infrastructures include (1) electricity and water networks [15,16], (2) electricity and transportation networks [17,18], (3) natural gas and electricity networks [19,20], (4) heating and electricity networks [21], among others.

To enhance planning, maintenance and emergency decision making, efficient and effective modeling and simulation tools are required. In this regard, Ouyang provides a comprehensive review of a variety of modeling frameworks for interdependent systems in [22]. In particular, graph theory based modeling method is one of the most frequently employed techniques, particularly to understand and quantify the coupling phenomena between different infrastructures [14,

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23]. Graph theory has formed the basis of many research works for network based systems, such as transportation systems [24], power grids [25], water networks [26], supply chains [27], and healthcare systems [28]. Conventionally, nodes of a graph representation denote the various infrastructure or service components and edges signify the physical connections among them. To exploit the conventional graph theory based approaches to model interdependent infrastructures, the network science community introduced the concept of “multilayer network”. This approach can facilitate modeling of “systems of systems” that crosses entity boundaries. Several prior publications have presented the relevant research works in this field [29–33]. For instance, in [29], the topological characterization of two interdependent small-sized networks is analyzed using multi-layer network concept to investigate the effects of interdependency on network recovery capability. Similarly, Yu et al. address the issue of uncertainties in edge connections across interdependent critical infrastructures via establishing the stochastic block models [30]. In [31,32], a novel graph model to map the complex interactions among infrastructures and extract interdependencies among the building systems, considering shocks and stresses, is proposed and explored. In [33], a graph based modeling approach is developed to analyze topological factors in a coupled power-law network to characterize network robustness, considering dependency types among networks. However, the conventional underlying modeling philosophy is adopted in the relevant works mentioned earlier, i.e. nodes represent resources and edges denote their physical connections. Although the conventional graph representation of interdependent infrastructure networks can reveal the topological information and fundamental relationship among system components, for complex systems, it does not provide in-depth knowledge about system processes and mutual dependencies among resources. In other words, questions such as “What are the dependencies of a particular facility on the others?”, “What processes are performed within nodes and edges?”, “How important are nodes and edges for system performance?” cannot be properly addressed by analyzing conventional graph representations [34]. Therefore, Farid et al. [14] introduce a “Hetero-functional Graph Theory” (HFGT) framework to rigorously model interdependent infrastructure networks and tackle all the challenges unaddressed within the conventional graph theory paradigm. The HFGT framework has developed from roots in the Axiomatic Design for Large Flexible Engineering Systems, which provides a rigorous platform for modeling systems of systems [35]. This newly proposed graph modeling philosophy is further exploited and extended in this work.

In this paper, for the first time, interdependent urban infrastructure networks (IUNs) including electricity, natural gas, heating, water and road transportation are modeled under a proposed weighted HFGT (WHFGT) framework to investigate their robustness against contingencies. Since recovery is considered as a vital aspect of resilience [36], relevant repair facilities for post-fault recovery after damage are also incorporated into the modeling framework. These facilities could be fire departments, maintenance companies, among others. The proposed WHFGT framework is used to evaluate the robustness of the coupled complex system. WHFGT is based on HFGT framework [14], which can be viewed as an intellectual fusion of model-based systems engineering and network science, and has been applied in transportation networks [37], production systems [38], power grids [39], among others. It aims to merge the fields of system engineering and network science for holistic modeling of interdependent engineering systems. Primarily, it is comprised of seven graph-based models that are related to their counterparts in model based system engineering. These graph-based models include (1) *System concept*; (2) *Hetero-functional adjacency graph*; (3) *The controller agency graph*; (4) *Controller adjacency graph*; (5) *Service as operand behavior*; (6) *Service feasibility graph*; (7) *System adjacency matrix*. The first two models are structural models and can be applied to many types of engineering systems. Due to explicit differentiation of system functionalities in the system concept model, the networks with unlike functions can be integrated into a single mathematical

model via models (1) and (2). Unlike the conventional graphic representation, where nodes and edges represent system elements, nodes of the hetero-functional graph represent system functionalities and edges represent the logical sequences or dependencies among them. The models (3) and (4) constitute the system control models. Together, they differentiate systems based upon the structure of their control and decision-making architectures. The models (5) and (6) constitute the service models, which differentiate systems based upon the behaviors of their operands. These models are then ultimately coupled together for a holistic modeling of interdependent engineering systems in a matrix form that is defined as the model (7) (i.e. system adjacency matrix). On the one hand, from the systems engineering perspective, HFGT can provide quantification of several structural concepts found in model-based systems engineering languages. From the network science perspective, it presents an alternative approach to construct graphs with fundamentally different meanings and insights.

In this paper, we focus on the System concept model (model 1), as the modeling of physical resources and their dependency relations is the key. To further elaborate model (1), an example is provided here. Considering three resources in a urban setting, namely a thermal power plant, a water treatment plant and a residential community that consumes treated water, the entire procedure can be described as the power plant generates electricity that is transmitted to the water treatment plant via power lines. Consequently, the water treatment plant purifies the raw water that is delivered to residential consumers via water pipelines. For this situation, using the conventional multi-layer graphic representation, the facilities of thermal power plant, water treatment plant and residential consumer are denoted as nodes in corresponding layers representing electric power and water networks. The edges between the resources represent their physical connections, i.e. power lines and water pipelines. On the other hand, in the system concept model of HFGT framework, the nodes represent the functionalities that are present in the system and the edges denote the dependencies among these functionalities. In this example, the functionalities include “generating electricity in thermal power plant”, “delivering electricity from thermal power plant to water treatment plant”, “purifying raw water at water treatment plant”, “delivering water to residential community” and “consumed treated water in residential community”. Consequently, the nodes of the HFGT model represent these functionalities and edges signify their dependencies. For instance, as the node corresponding to “consuming water in residential community” relies on the node corresponding to “delivering water to residential community”, an edge is present between these nodes, whose weight quantifies the degree of dependency. As seen, the HFGT based graphic model can reveal in-depth information about functionalities and dependencies of system components in a convenient manner. In other words, HFGT can explicitly model complex relationships among relevant functionalities of physical entities, while the multi-layer network approach is limited by the modeling of physical configurations.

In this work, we propose to adopt WHFGT to model an IUN, comprising of an electrical power system, a water network, a district heating network, a natural gas system and a road transportation network for delivering repair service, which extends the pioneer works on HFGT [14,35,39] to further accommodate more generic situations. Specifically, in contrast to binary weights of edges, which can merely indicate if the dependency relation is present or not, we propose to incorporate weights with real numbers for edges to quantify the dependency degradation, namely WHFGT framework. Edge weights with real numbers are of great importance to examine robustness of a complex network, as effects of contingency severity can be integrated. Furthermore, the network related constraints of involved IUN are considered here. As a result, the robustness of IUN is assessed via conducting various attack simulations on the proposed WHFGT model. It is worth to note that the “attack” word used throughout this paper, represents contingencies (complete or partial loss of nodes) in the network that can be caused by both malicious attacks and natural disasters.

1.3. Contributions

The major contributions of this paper are listed below.

- We present a WHFGT based framework to provide a comprehensive and in-depth robustness evaluation of an IUN. The proposed WHFGT modeling paradigm has a different definition philosophy of nodes and edges, compared to the conventional graphic representation, which aid in revealing the various functionalities within the network and dependencies among them. The dependencies are determined by the procedures of different functionalities and network related constraints.
- Unlike [14] that uses binary terms to quantify dependencies, this paper proposes edge weights with real numbers in the WHFGT framework to quantify dependencies. This improvement offers a more generic framework, which enables the simulations of partial attacks (partial loss of functionalities due to attacks). With the proposed WHFGT framework, we conduct an in-depth analysis about the IUN that reveals system robustness against disruptions from multiple dimensions.

The remainder of this paper is organized as follows. Section 2 presents an overview of conventional and HFGT models with proposed WHFGT framework. Section 3 discusses robustness evaluation and various attack strategies. In Section 4, numerical results are presented along with the comparison across attack strategies. Section 5 presents the conclusions and future research suggestions.

2. Modeling integrated urban networks

This section presents existing paradigms for modeling IUN, followed by an overview of the proposed framework. Specifically, the system knowledge base of HFGT that maps the allocation of system resources to system processes will be depicted and elaborated here.

2.1. Conventional graph theory and hetero-functional graph theory (HFGT) framework

A conceptual diagram for an IUN is depicted in Fig. 1, which is used as an example to illustrate the interactions among infrastructures. In this directed graph representation, the nodes denote various infrastructures and facilities, while directed edges represent the energy/water/service flow among the nodes. As seen, networks of electricity, natural gas, heating, and water are intertwined to support each other. For instance, the operative gas-driven power plant (PP1) depends on the gas supply from the gas source (GS) and water supply from the water treatment plant (WTP). Besides, the effective maintenance and repair service for PP1 relies on the power infrastructure repair facility (RP). The repair facility is responsible for post fault recovery after damage. These facilities could be fire departments, maintenance companies, among others. On the other hand, PP1 provides electricity to other infrastructures and facilities, such as GS, WTP, among others. Additionally, customer demands (CD) of energy and water supplies are supported by the corresponding facilities.

As mentioned earlier, although the conventional graph representation can reveal the topological information and fundamental relationship among system components, for complex systems, it does not provide in-depth knowledge about system processes and mutual dependencies among resources. Therefore, Farid et al. introduce an HFGT framework to rigorously model IUN and tackle all the challenges unaddressed within the conventional graph theory paradigm. Essentially, HFGT framework is comprised of seven graph-based models, whose detailed descriptions are provided earlier. Fig. 2 illustrates a summary of key elements and functions of these models.

In this work, we focus on the system concept module to model IUNs and conduct robustness analysis. As seen from Fig. 2, system functionality is the key element, which is defined as an activity that

transforms a predefined set of inputs into a predefined set of outputs, e.g. generating electricity from a PP, producing treated water from a WTP, among others. System resource is a physical entity which executes the processes, such as PP, WTP, GS, etc. The system concept model provides a rigorous platform to conduct this transformation. Specifically, the system knowledge matrix (J_S) in the system concept model is a binary matrix, which maps the processes (P) with the corresponding resources (R) as follows:

$$P = J_S \odot R \quad (1)$$

where $\odot$ is matrix boolean multiplication [40]. The element in the J_S that projects the element in R to the corresponding element in P is equal to unity, when an action that converts the resource to the process exists. For instance, a power plant, as a system resource, can be mapped to generating electricity, as a system functionality, via the knowledge matrix (J_S). Accordingly, the particular element in J_S representing the electricity generation action being executed by the power plant is set to unity.

2.2. Weighted hetero functional graph theory (WHFGT) framework

Using the knowledge matrix of the HFGT framework, system resources and system processes can be mapped. To more realistically capture functionality interactions, we propose a process relation matrix (P_R), which is developed on top of the conventional dependency matrix of HFGTs. For the conventional dependency matrix, the relationships between the functionalities are expressed in binary terms, i.e., it only captures whether the dependency is present or not. However, it is inadequate to quantify the degree of dependency between the two functionalities. For instance, an electric power consumer can be supplied by multiple power plants. The conventional HFGT framework cannot present the degree of dependency of the electricity consumer on each power plant precisely. Therefore, to address this shortcoming, we propose a WHFGT framework and the corresponding relation matrix (P_R), which quantifies the degree of dependency using weights of real numbers. As performing a particular functionality could depend on multiple other functionalities, the proportion contributed from each functionality signifies the weight. Thus, for a particular node in the WHFGT, the weights of its associated edges in matrix P_R denote the degree of dependencies on other functionalities. For instance, in Fig. 1, electricity demand of the community customers can be supplied by two sources, i.e., the gas-driven power plant and CHP, and the dependencies on both of them can be reflected by assigning weights. With the assumption that gas-driven plant contributes 20% of the total demand and CHP supplies remaining 80%, we can allocate 0.2 to the edge connecting the functionality of “consuming electricity in community” and the functionality of “delivering electricity from gas-driven power plant to community consumer”, and 0.8 to the other edge. For the illustration purpose, we have arbitrarily allocated weights in this work. Note that the proposed method can be adapted to accommodate other parameters in practice based on the collected relevant information. The process relation matrix (P_R) for the conceptual diagram depicted in Fig. 1 is a 43×43 matrix as 43 functionalities are incorporated. For brevity, the process relation matrix (P_R) is provided in an online repository [41]. The rows and columns of the matrix represent functionalities, while the entries represent the dependencies between two corresponding functionalities. Correspondingly, Fig. 3 depicts the process relation graph for a synthetic network shown in Fig. 1. The direction of edges are from the source to target nodes, where target nodes (functionalities) are dependent on the source nodes (functionalities) with dependency proportional to the edge weights. In the following sections, extensive analysis will be conducted on the process relation graph to evaluate robustness of IUN. The WHFGT allows one to study the interdependency in a more informative way in a sense that the dependency weights could play an important role in various graph robustness analysis (percolation, cascading failures, etc.).

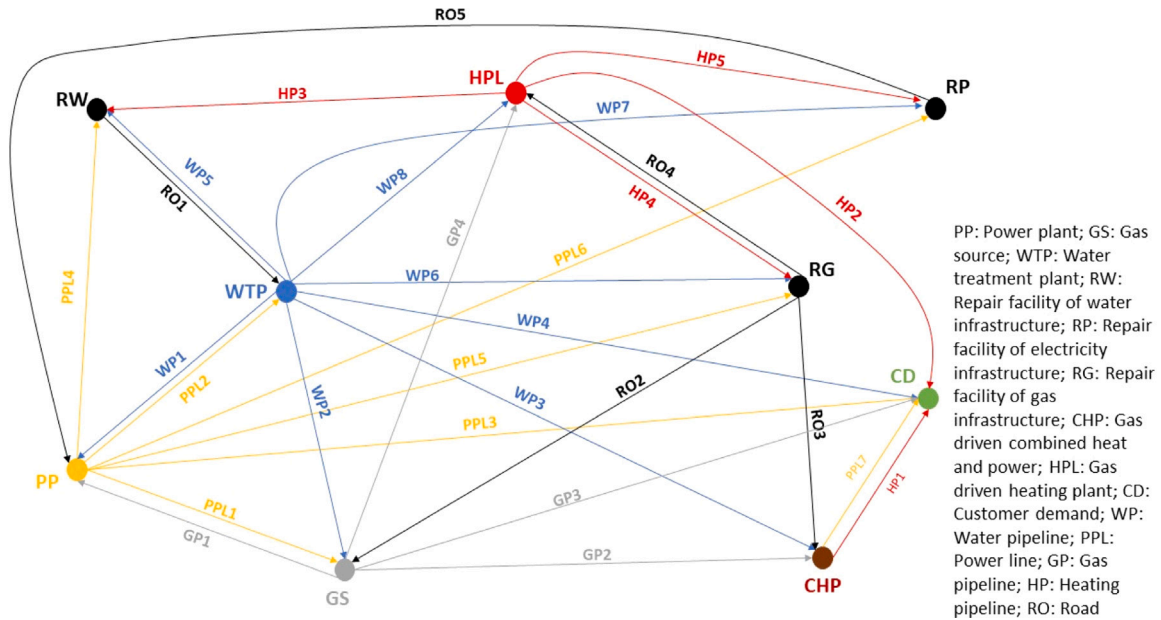


Fig. 1. Conventional Graph Representation of a Conceptual Interdependent Urban Infrastructure Network (IUN).

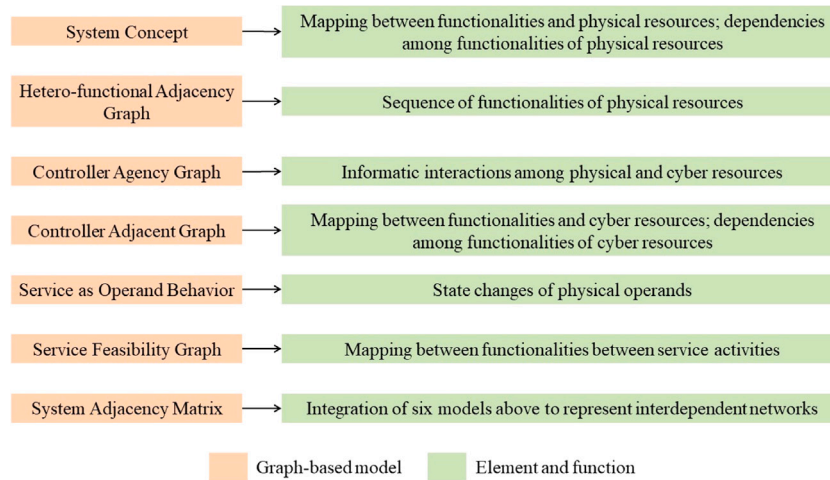


Fig. 2. Illustrations of models within HFGT framework.

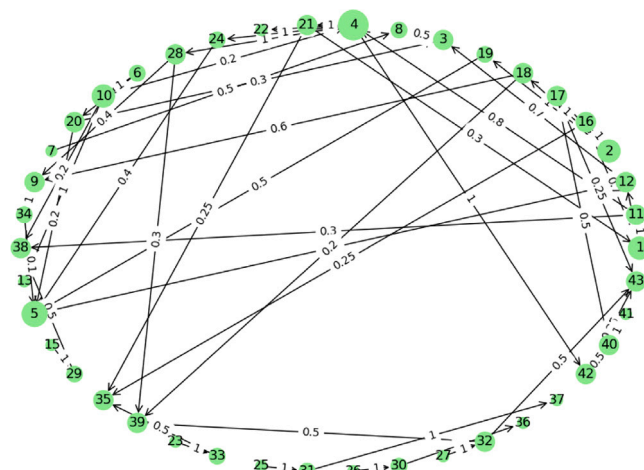


Fig. 3. Process relation graph corresponding to the network shown in Fig. 1.

3. Contingency analysis in WHFGT framework

The tight interconnections among various infrastructures may exacerbate the impact of potential cascading failures, as contingencies in one part of the system might propagate to other networks. Therefore, robustness evaluation of IUNs is critical for vulnerability analysis and robustness enhancement.

To simulate contingencies on IUNs, the nodes are defined as the targets for attacks. In other words, disabling certain functionalities is equivalent to removing the associated nodes in the WHFGT graphs. Note that the edges merely denote the dependencies among functionalities, thus, they are not considered as attack targets, as disabling dependency relations has no interpretative meaning. Besides, based on graph percolation procedure [42], nodes are removed in stages and the robustness metrics of the residual graph are computed at each stage to study the transition properties of the graph. Along with the removal of a selected node, the child nodes of the selected node are also removed under specific conditions. Here, child nodes are defined as nodes that can be reached (directed path exist) from the selected node and are computed using depth first search algorithm [43]. We would like to reiterate that, for the sake of simplicity, all the contingencies in this paper are referred to as “attacks”.

According to the available system information and attacker capabilities/strength, the attacks can be further divided into four categories, as tabulated in Table 1. Random/targeted attacks are categorized based on the knowledge level of attackers about the system. Specifically, in a random attack, nodes are removed randomly at each stage of the percolation whereas, in a targeted attack, nodes are removed based on certain importance scores of nodes (assuming the system information is available to the attacker). On the other hand, complete/partial attacks are categorized based on attackers capability. Specifically, in a complete attack, the attacker disables a process completely, while in a partial attack, the attack can only degrade the process to a certain degree. Detailed description of these attack categories will be provided later.

3.1. Graph centrality

As mentioned earlier, targeted attacks are launched based on the importance scores assigned to nodes, assuming the attacker has acquired the entire system information. Here, the concept of centrality is applied to quantify importance of nodes in a network. Four types of centrality are widely applied in prior literature, including degree centrality, eigenvector centrality, PageRank centrality and betweenness centrality [44]. The degree centrality indicates the number of connected edges for each node. The intuitive interpretation of degree is that nodes with more connections imply involvement in more functionalities and transformations. Moreover, for directed graphs, degree centrality can be separated into in-degree and out-degree, which represent the number of incoming edges and outgoing edges, respectively. Thus, weighted out-degree denotes the total sum of outgoing edge weights. Eigenvector centrality provides an alternative perspective of the node's importance. Here, the focus is on quantifying the influence of connected nodes, instead of using the number of connected nodes. A potential problem faced by eigenvector centrality is that if one node is among a large number of nodes connected to another node with substantially high centrality, the eigenvector centrality index would overestimate the influence of that particular node. PageRank centrality is proven to be able to resolve this issue and is one of the most well-known and frequently used measure in search engines [45]. Finally, the betweenness centrality measures the extent to which a node lies on the shortest paths between other nodes. Nodes with high betweenness are expected to have considerable influence within a network by virtue of their control over information passing between others. In IUN, we select betweenness and weighted out-degree centrality as the basis to conduct targeted attacks. Specifically, removal of nodes with high betweenness

Table 1

Types of attack strategies.

Strength	Information	
	Random	Target
Complete	Complete random	Complete target
Partial	Partial random	Partial target

centrality and high weighted out-degree centrality in the earlier stages is considered as efficient approaches to impair the system connectivity, thereby simulating extreme conditions. While betweenness centrality and out-degree centrality are selected as an example to implement targeted attacks in this paper, the study can be extended to other guiding metrics as well.

3.2. Complete attack

The most trivial type of attack scenario would be the situation where the attacker is capable of disabling a process completely, referred to as a complete attack. This can be caused by extreme natural events or powerful malicious attackers. In WHFGT, complete attacks can be represented by removal of corresponding nodes entirely, within a single stage. Based on the knowledge level of the attacker, complete attacks can be further categorized into complete random and complete targeted attacks. Specifically, for a complete random attack, it is assumed that the attacker is not aware of the internal configuration of the IUN and hence the targeted nodes (processes) will be attacked randomly. Under this situation, nodes are selected randomly for removal at each stage of the percolation and the percolation is continued until the graph is left with two nodes. Obviously, if a child node (functionality) is dependent on its only parent node (functionality), disabling the parent node (functionality) will lead to obliteration of the child node. For instance, the node representing the functionality of “transport water from WTP to the residential community” will be removed once the node corresponding to the functionality of “treat water in WTP” is out of the graph. On the other hand, for child nodes with multiple parent nodes, removing some (not all) of its parent nodes will lead to a certain degree of degradation, which is quantified by the out-degree weight reduction. Out-degree weights of a node indicate the amount of dependency the neighbors have on the node, whereas the in-degree weights of a node signify the extent of dependency the node has on its neighbor. Specifically, the decrease of out-degree weights is assumed to be in proportion to the decrease of in-degree weights. In other words, for a particular node (process), with less inputs from parent nodes (functionalities), its performance of supporting other nodes (functionalities) will be degraded accordingly. When the out-degree weight of a node decreases to a pre-defined threshold, namely critical quality of functionality (QoF), it will be removed from the graph. The practical interpretation of this rule is that when a process is unable to support certain follow-up services, it will be considered as dysfunctional. For instance, when the process of “deliver electricity to customers 1–5” is degraded to “deliver electricity to only customer 1” due to attacks, the relevant infrastructure might be considered as dysfunctional and the temporary shutdown or maintenance actions may be needed. Note that the critical QoF value can be assigned flexibly, based on the specific processes and situations.

Furthermore, considering the frequent occurrences of malicious cyber attacks around the globe, it is increasingly vital to address the situation wherein the attacker could obtain insights on the entire system and launch targeted attacks. Targeted attacks aim to remove nodes with higher weighted out-degree/betweenness centrality in the earlier stages of percolation. In other words, functionalities with higher importance are targeted in the earlier stages of percolation. The complete targeted attack is similar to its random counterpart except that the nodes are selected on the basis of predefined importance score, determined by weighted out-degree/betweenness centrality. During targeted attacks,

Algorithm 1 Complete attack

Input: Graph G with V nodes.
Output: List of robustness metrics corresponding to various percolation stages
LOOP Process

```

1: while  $|V| > 2$  do
2:   nodeselectd: Select a node randomly or based on the order of
     important scores.
3:   childnodes: Find all child nodes of nodeselectd.
4:   remove nodeselectd from  $G$ 
5:   for child in childnodes do
6:     indeg: Find weighted in degree of child node
7:     outdeg: Find weighted out degree of child node
8:      $Degradationratio = \frac{\text{current indeg of child}}{\text{indeg of child before percolation}}$ 
9:     outdegweights: Decrease the out edges weight of child by
       Degradationratio
10:    if (indeg = 0) or (outdegweights < Quality of Function) then
11:      remove child from  $G$ 
12:    end if
13:  end for
14:  compute robustness metrics
15: end while
16: return list of robustness metrics corresponding to various
    percolation stages

```

nodes are removed in descending order of importance. The detailed procedure of complete attack is shown in Algorithm 1. The robustness metrics indicated in the output of Algorithm 1 will be elaborated in Section 4. Again, it should be noted that the word “attack” used in this paper does not necessarily denote malicious cyber/physical destruction, but also represents natural disasters that may lead to serious damage.

3.3. Partial attack

The complete attack is designed for simulating the scenarios where the attack is sufficiently powerful to destroy a functionality completely, which results in loss of the entire process. To examine the robustness of the graph against less severe contingencies, we propose a new type of attack strategy referred to as partial attack. The key difference between partial and complete attack strategies is that during percolation of a partial attack, the attacked nodes will only experience out-degree depletion until the critical QoF value is violated. In contrast to complete attacks where the attacked process is disabled unconditionally, partial attacks will only lead to a certain level of degradation, namely degradation level, which signifies the extent of performance degradation of functionalities. In other words, functionalities operating at the level above QoF value is considered to be at an operative stage. Once degradation level of a functionality reaches QoF threshold, it becomes no longer operative that results in the removal of corresponding node from the graph. On the other hand, similar to complete attacks, partial attacks can be further subdivided into two types depending upon the knowledge level of the attacker, namely partial random attacks and partial targeted attacks. For partial random attacks, the attacked nodes are selected randomly, while nodes with higher weighted out-degree/betweenness centrality will be targeted in the early stages during partial targeted attacks. Algorithm 2 describes the complete procedure of a partial attack. It should be noted that partial attack analysis only applies to our proposed weighted hetero functional graphs, since partial degradation of weights can be incorporated. Traditional HFGT framework cannot accommodate these realistic attack scenarios.

Algorithm 2 Partial attack

Input: Graph G with V nodes.
Output: List of robustness metrics corresponding to various percolation stages
LOOP Process

```

1: while  $|V| > 2$  do
2:   nodeselectd: Select a node randomly or based on the order of
     important scores.
3:   childnodes: Find all child nodes of nodeselectd.
4:   outdegweights: Decrease the out edges weight of nodeselectd by
     random quantity.
5:   if (outdegweights < Acceptable service) then
6:     remove nodeselectd from  $G$ 
7:   end if
8:   for child in childnodes do
9:     indeg: Find weighted in degree of child node
10:    outdeg: Find weighted out degree of child node
11:     $Degradationratio = \frac{\text{current indeg of child}}{\text{indeg of child before percolation}}$ 
12:    outdegweights: Decrease the out edges weight of child by
      Degradationratio
13:    if (indeg = 0) or (outdegweights < Quality of Function) then
14:      remove child from  $G$ 
15:    end if
16:  end for
17:  compute robustness metrics
18: end while
19: return list of robustness metrics corresponding to various
    percolation stages

```

4. Numerical study

In this section, we implement WHFGT based modeling of IUN and evaluate its robustness. To begin with, the system network configuration is described, along with its effects on the process relation matrix (P_R). After introducing several robustness metrics, the IUN robustness is quantified by analyzing the impacts of four types of attack scenarios. Finally, the application of proposed robustness assessment method in optimizing dependency relations among infrastructures for robustness enhancement is explored.

4.1. System network configuration

Fig. 4 shows the integrated urban network of interest. This network comprises of an electric power system [46], a gas network [47], a water system [48], a road transportation network [49], and a district heating network [50]. Further, Table 2 tabulates the locations of various infrastructure facilities within the networks, including 1 gas-fired power plant (GPP), 1 solar power plant (SPP), 2 water treatment plants (WTP), 3 heating plants (HPL), 2 gas stations (GS), 1 gas-driven CHP, 2 industry parks (IP1-IP2), 3 commercial zones (CZ1-CZ3), 6 residential zones (RZ1-RZ6), 1 water infrastructure repair facility (WIRF), 1 power infrastructure repair facility (PIRF) and 1 natural gas infrastructure repair facility (GIRF). To ensure the proper operation of these facilities, 121 functionalities are required in total. In particular, electricity supplies of industry parks, commercial zones and one residential zone (RZ1) depend on three electric power sources, including the gas-driven power plant, CHP and the solar plant. The complete process relation matrix is presented in [41], with the same structure as defined earlier for the conceptual diagram with 43 functionalities.

For this system network configuration, the relation matrix needs to be extended to incorporate the additional dependencies among processes due to the shared path of energy/service deliveries. For instance, the successful execution of the process that transmits power from GPP (located at node 5 in electric power network) to WIRF (located at node

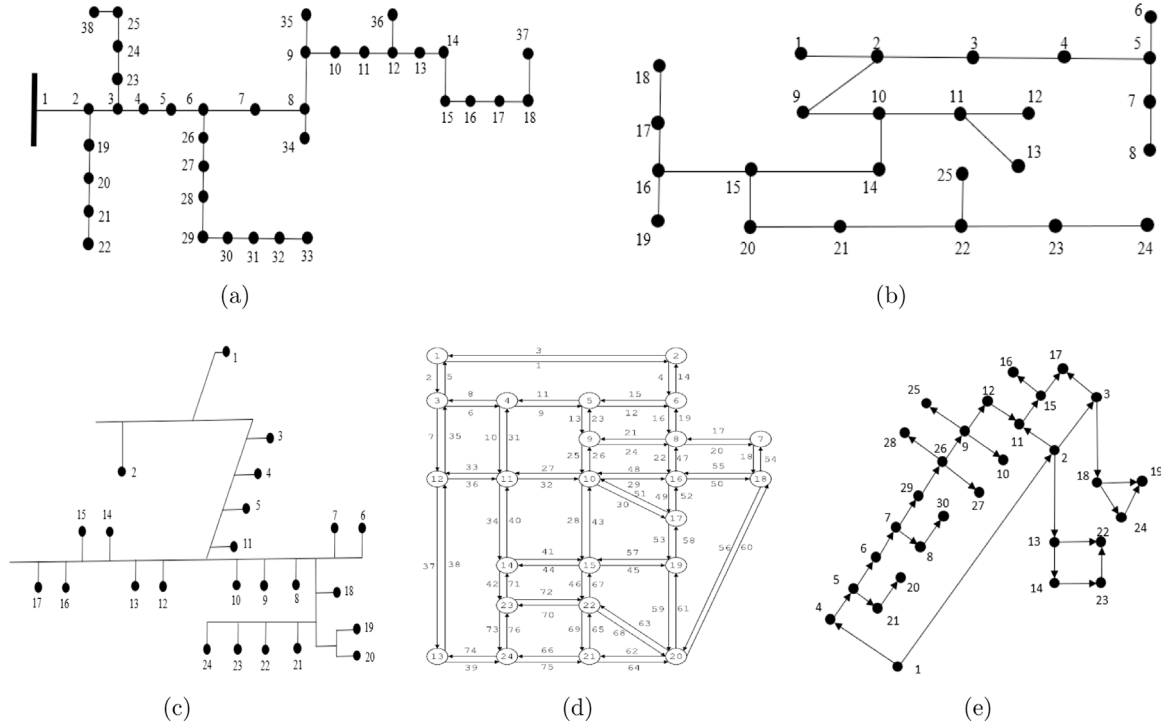


Fig. 4. Integrated infrastructure with (a)electricity, (b)gas, (c)district heating, (d)transportation, (e)water networks.

1 in electric power network) depends on the successful execution of process that transmits power from GPP (located at node 5 in electric power network) to WTP1 (located at node 3 in electric power network). This is because, the electricity delivery path between GPP (node 5) and WTP1 (node 3) is a part of the only path for electricity transmission from GPP (node 5) to WIRF (node 1). Similarly, the effects of other network constraints on process dependencies are integrated into the process relation matrix.

4.2. Metrics for robustness evaluation

Robustness of IUN in the present work is studied through the percolation process, where nodes of the network are removed in stages and the connectivity of the residual network is assessed at each stage, until the graph is completely disconnected. In a high level definition, the connectivity of the residual graph after node removal represents the robustness of the graph towards that node attack. In general, higher connectivity of the residual graph indicates enhanced robustness. The conventional indices for quantification of graph connectivity include Largest Connected Component (LCC) and the Number of Connected Components (NCC) [51]. Here, a component denotes a subgraph, in which any two vertices are connected to each other by paths and is connected to no additional vertices in the graph. Specifically, LCC denotes the size of the largest component where every node is at least connected to one other node, while NCC represents the number of connected components in the entire graph. Furthermore, we use flow robustness (FR) to quantify robustness, from the components standpoint [52]. It captures the ability of the nodes to communicate with each other in all the clusters and hence characterizes the overall reachability of the graph. Unlike LCC that only accounts for the largest connected component, FR incorporates the number of nodes in all components of the residual graph. The FR metric corresponds to:

$$FR = \frac{\sum_i |C_i|(|C_i| - 1)}{N(N - 1)} \quad (2)$$

where C_i is the number of nodes in component i and N denotes the total number of nodes in the original graph before attacks. As

seen, FR represents the degradation at a global level by monitoring the connectivity situation of all components and hence reveals global robustness.

However, the three indices described earlier are inherently incapable of incorporating weights of edges. Therefore, we design a new robustness metric, namely service robustness (SR). It can be expressed as:

$$SR = \frac{\sum_i Q_i^a}{\sum_i Q_i} \quad (3)$$

where Q_i^a denotes the weighted out-degree of node i after each stage of attack and Q_i represents the total weighted out-degree of all nodes in the original graph without attacks. The SR index not only infers connectivity of the graph, but also incorporates the weights of edges. Since the weights of edges quantify the dependencies between source and target nodes, integrating them into the robustness analysis would aid in providing a more precise evaluation of impacts of attacks.

To summarize, it is obvious that higher values of LCC, FR and SR indicate a more robust network, as they quantify reachability and connectivity from various dimensions. However, the relationship between the NCC and robustness is not straightforward. Specifically, at the initial stages of decomposition process of the graph, the value of NCC may increase since the entire graph is decomposed into several sub-graphs (components). After a certain stage, the value of NCC decreases gradually, as components start to vanish into individual nodes. In the next sub-section, we examine the robustness of the graph described earlier by observing the trajectories of the four robustness metrics along the percolation stages with different types of attacks.

4.3. IUN robustness analysis

The robustness of IUN is examined by evaluating the robustness metrics, i.e., LCC , NCC , SR and FR , after each stage of percolation. The impacts of four attack strategies, i.e., complete random attack, complete targeted attack, partial random attack and partial targeted attack, are examined and compared in the following. The attacked node selections in targeted attack are carried out based on betweenness and

Table 2
Locations of entities in various networks.

Utility	Description					
	Entity	Location	Entity	Location	Entity	Location
Power	WIRF	Node 1	CZ1	Node 9	RZ2	Node 17
	WTP1	Node 3	WTP2	Node 11	RZ3	Node 19
	IP1	Node 4	CZ2	Node 12	RZ4	Node 20
	GPP	Node 5	RZ1	Node 13	GS2	Node 21
	IP2	Node 6	GIRF	Node 14	RZ5	Node 22
	GS1	Node 7	SPP	Node 15	CHP1	Node 23
	PIRF	Node 8	CZ3	Node 16	RZ6	Node 24
Water	WIRF	Node 1	CZ1	Node 9	RZ2	Node 17
	HPL1	Node 2	HPL2	Node 10	HPL3	Node 18
	WTP1	Node 3	WTP2	Node 11	RZ3	Node 19
	IP1	Node 4	CZ2	Node 12	RZ4	Node 20
	GPP	Node 5	RZ1	Node 13	GS2	Node 21
	IP2	Node 6	GIRF	Node 14	RZ5	Node 22
	GS1	Node 7	SPP	Node 15	RZ6	Node 24
Gas	PIRF	Node 8	CZ3	Node 16		
	GS1	Node 1	HPL3	Node 8	RZ3	Node 14
	HPL1	Node 2	CHP1	Node 10	RZ4	Node 15
	GPP	Node 5	RZ1	Node 12	RZ5	Node 17
Heat	HPL2	Node 6	RZ2	Node 13	RZ6	Node 19
	WIRF	Node 1	CZ2	Node 12	RZ3	Node 19
	IP1	Node 4	RZ1	Node 13	RZ4	Node 20
	IP2	Node 6	GIRF	Node 14	RZ5	Node 22
	PIRF	Node 8	CZ3	Node 16	CHP1	Node 23
Transport	CZ1	Node 9	RZ2	Node 17	RZ6	Node 24
	WIRF	Node 1	PIRF	Node 8	HPL3	Node 18
	HPL1	Node 2	HPL2	Node 10	GS2	Node 21
	WTP1	Node 3	WTP2	Node 11	CHP1	Node 23
	GPP	Node 5	GIRF	Node 14		
	GS1	Node 7	SPP	Node 15		

weighted out-degree. Additionally, Monte-Carlo simulation approach is used here to evaluate the effects of random attack strategies on robustness metrics. Here, the random selections of attacked nodes are repeated 10,000 times.

4.3.1. Complete attack

As mentioned earlier, in a complete attack, the attacked nodes are removed completely once the attack is imposed. Under this category, we first simulate the random attack situation. The trajectories of the four robustness metrics are depicted in Fig. 5. The horizontal axis denotes the numbers of sequential attacks imposed on the system. After each attack, the robustness metrics of the residual graph are evaluated and plotted in the vertical axis. The robustness analysis is performed in steps, namely in percolation stages, which is in consistent with the definition of percolation in [53]. In this way, real-world situations such as sequential cyber-attack events or natural disasters can be simulated. As expected, LCC decreases along with the progress of percolation, as node removals will decompose the graph into various components with a smaller number of nodes. On the other hand, the NCC increases in the initial stages and starts declining after a certain stage. This is because, initially, the entire graph is fragmented into several sub-graphs that lead to an increasing number of connected components. After a certain stage (stage 14 in this case), those sub-graphs will be further decomposed into individual nodes, decreasing the value of NCC. The global metric FR also drops exponentially but with a lower decay rate compared to LCC, which implies the slower degradation of connectivity (weighted or unweighted) among various components across the entire graph (compared to the local connectivity in the largest component). The trajectory of SR shows a gradual decline as well, which captures the diminishing mutual dependencies among nodes.

To demonstrate the effects of attackers' knowledge about the system, we further illustrate the trajectories of robustness metrics with imposed betweenness- and weighted outdegree-based targeted attacks along the percolation stages, as depicted by blue and green curves in Fig. 5, respectively. Like the results associated with complete random

attacks, the general patterns of trajectories of robustness metrics correspond to the trend of gradual decline except for NCC. The NCC metric initially increases abruptly followed by fluctuations and eventually settles in the final stages. Additionally, the value of FR has a lower decay rate compared to LCC value, which indicates a certain level of connectivity among components even with few nodes in the largest component. Additionally, compared to a random attack, the degradation of the system robustness is substantially higher with targeted attacks. Specifically, it takes 12 and 10 stages to disconnect the graph entirely if betweenness and weighted out-degree are selected as the basis to launch targeted attacks, compared to an average of 40 stages required to completely disconnect the graph with imposed random attacks. To further quantify the distinctions between random and targeted attacks, the numbers of sequential attacks required for degrading the robustness metrics to certain levels are tabulated in Table 3. The results illustrate that targeted attacks require fewer steps to reach the same level of degradation as random attacks. For instance, achieving 50% degradation in FR only takes 3 and 1 stages for betweenness- and weighted outdegree-based targeted attacks, respectively, while 10 steps are needed to attain the same level of degradation with random attacks. In other words, this test demonstrates the value of securing system information, as attackers with system information are capable of causing severe destruction rapidly. Fig. 6 depicts the histograms of numbers of nodes in each component at various stages with a complete targeted attack. The rapid decrease of node number in each component at the early stages can be witnessed, while the component decomposition process slows down after stage 6. Furthermore, comparing two types of targeted attacks, we can observe that the weighted outdegree-based targeted attacks outperform betweenness-based targeted attacks for all robustness metrics. This implies that weighted outdegree centrality could provide a more precise indication of nodes importance in this test.

4.3.2. Partial attack

In partial random attacks, attacked nodes are selected randomly with a certain level of degradation at each stage of percolation. The

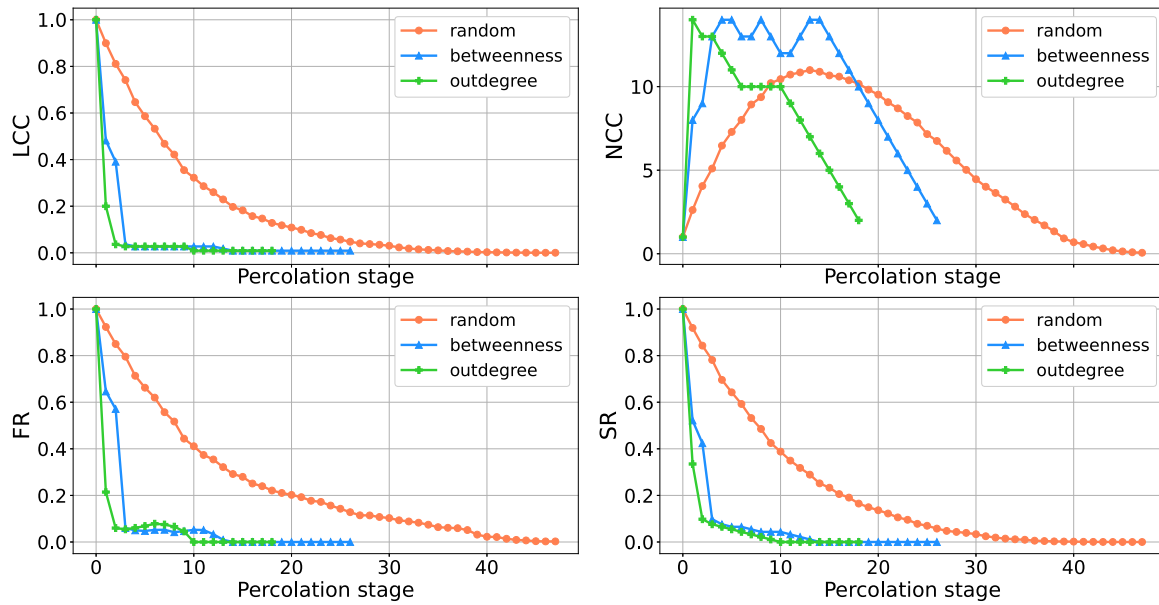


Fig. 5. Trajectory of robustness metrics across various stages of percolation with complete attack.

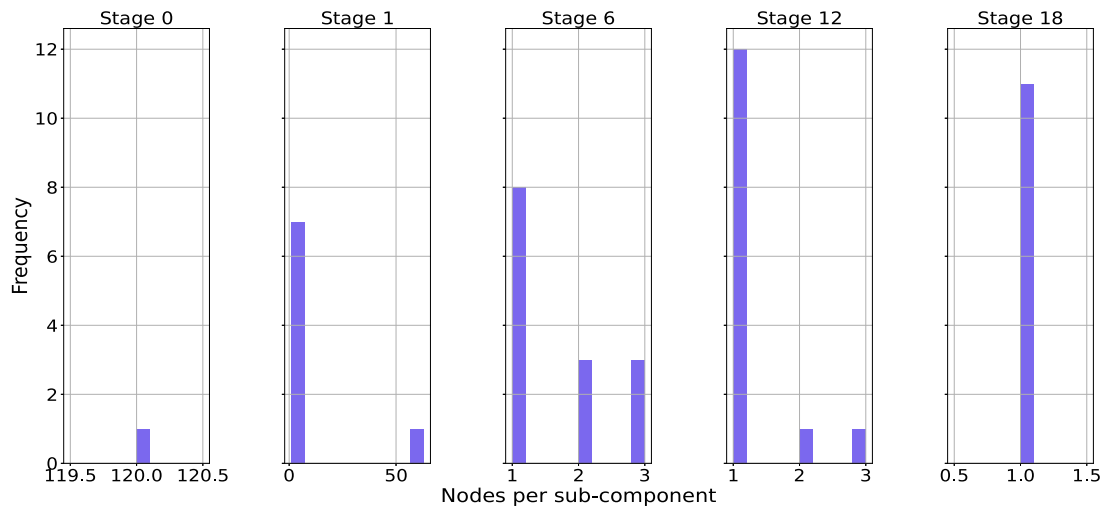


Fig. 6. Histogram of nodes per sub-component/cluster across different stages of percolation in complete targeted attack.

Table 3

Number of attacks for various degradation levels corresponding to random and targeted attacks with complete attack strategy.

Degradation level	20%			50%			80%		
	Attack								
Metric	RN	BW	OD	RN	BW	OD	RN	BW	OD
LCC	3	1	1	7	1	1	15	3	2
FR	4	1	1	10	3	1	22	3	2
SR	3	1	1	8	2	1	17	3	2

RN: Random attack BW: Betweenness based targeted attack OD: Out-degree based targeted attack.

orange curves in Fig. 7 depict the trajectories of robustness metrics. The general trends in Fig. 7 are similar to the results of a complete random attack. Specifically, the metrics of LCC, FR and SR decrease abruptly at the beginning followed by a moderate decline, while the index of NCC increases initially and decreases after a certain stage. These observations are consistent with the results from complete attack case described earlier.

We also illustrate the trajectories of robustness metrics along the percolation stages with partial targeted attacks, as depicted in blue and

green curves in Fig. 7. As seen in the figure, targeted attacks lead to a more abrupt decrease in LCC, FR and SR. To further illustrate the graph fragmentation process, Fig. 8 depicts the histograms of numbers of nodes in each component at representative stages of partial targeted attack. The rapid decrease of node number in each component at the beginning can be explicitly observed, while the component decomposition process significantly slows down after stage 10. The results correspond to the conclusion drawn from Fig. 7, which again signifies the remarkable ability of targeted attacks in decimating the network

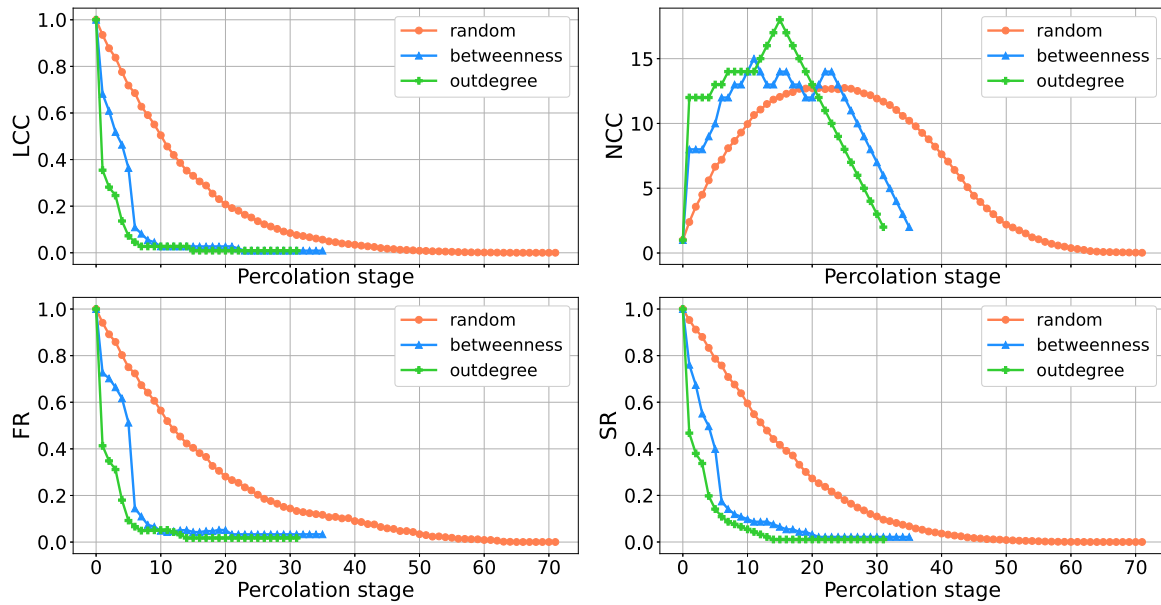


Fig. 7. Trajectory of robustness metrics across various stages of percolation with partial attack.

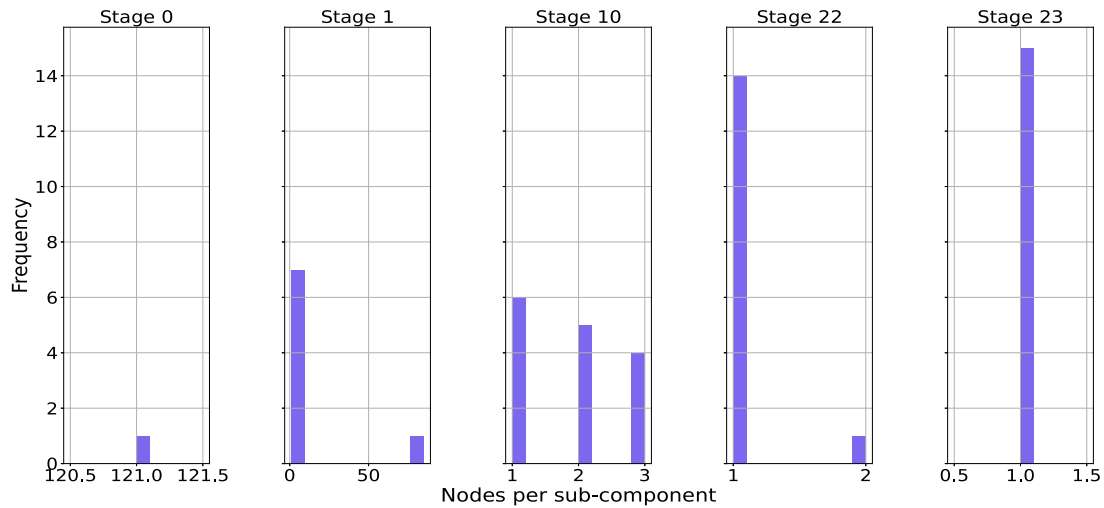


Fig. 8. Histogram of nodes per sub-component/cluster across different stages of percolation in partial targeted attack.

Table 4

Number of attacks for various degradation levels corresponding to random and targeted attacks with partial attack strategy.

Degradation level	20%			50%			80%		
Metric	Attack								
	RN	BW	OD	RN	BW	OD	RN	BW	OD
LCC	4	1	1	11	4	1	23	6	4
FR	4	1	1	13	6	1	30	6	4
SR	5	1	1	13	4	1	26	6	4

RN: Random attack BW: Betweenness based targeted attack OD: Out-degree based targeted attack.

rapidly. Additionally, the notable gap between partial random and targeted attack strategies can be observed in Table 4, which tabulates the required numbers of attack stages to reach certain degradation levels. As expected, it takes fewer steps for targeted attacks to fragment the graph into a certain level. In a nutshell, the general trend of robustness metrics are similar in both complete and partial attacks. The distinction between complete and partial attacks is compared with the distinction between targeted and random attacks in the following subsection.

4.3.3. Complete random attack versus partial targeted attack

Evidently, complete, targeted attacks cause more severe damage than partial, random attacks, as demonstrated in Tables 3 and 4. In this section, the comparison between the impacts of a complete random attack and a partial targeted attack is conducted to evaluate the merits of securing system information and infrastructure hardening.

From Tables 3 and 4, we can witness that, in general, the distinction between random and targeted attacks is more significant, compared to the distinction between complete and partial attacks. For instance, assuming the targeted attacks are based on out-degree weight, to reach

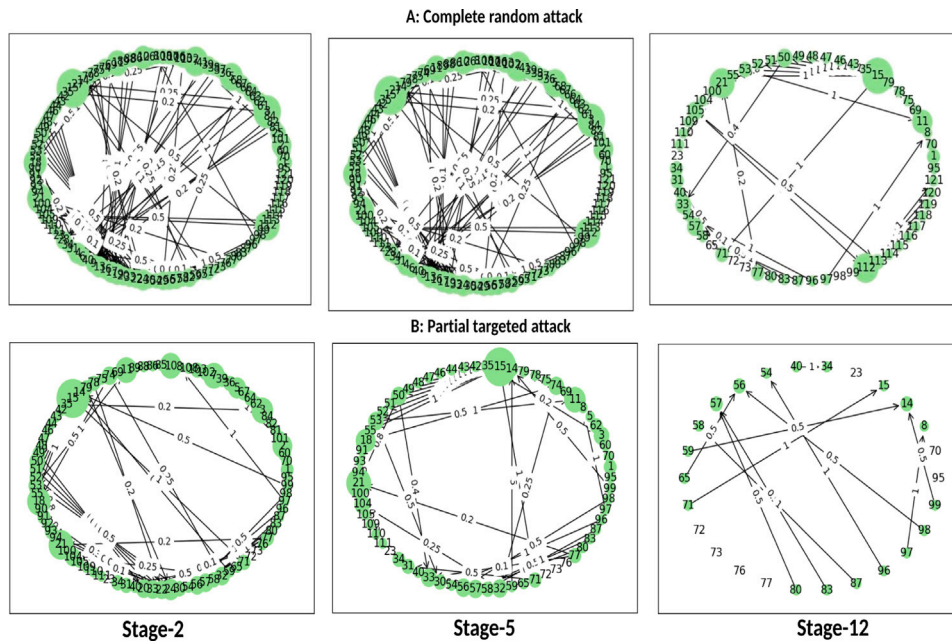


Fig. 9. Snapshots of graph across representative stages of percolation with complete random attack and partial targeted attack.

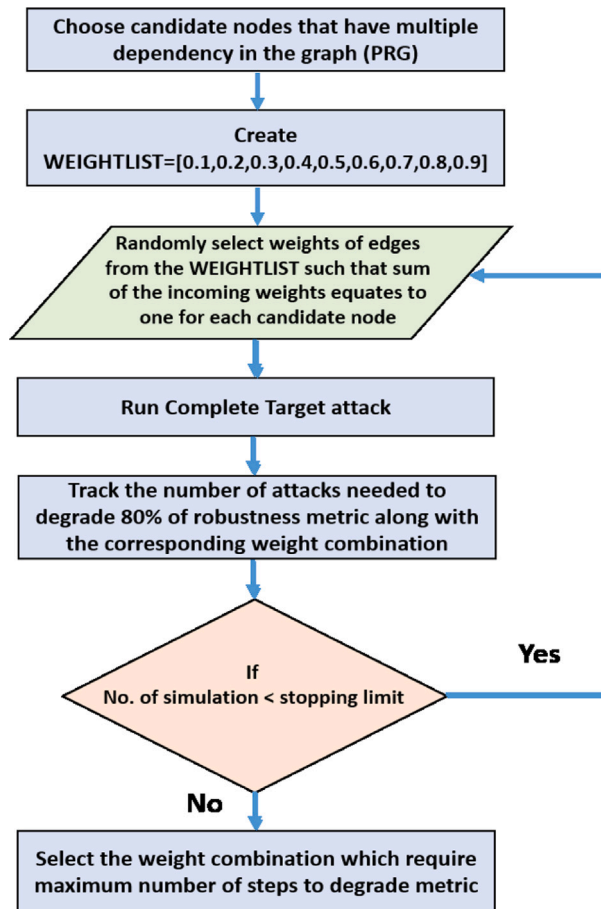


Fig. 10. Flowchart depicting the enumeration.

the degradation level of 80% in FR , the gap in the required percolation stages between a partial random attack and a complete random attack is 8, whereas this gap between a complete targeted attack and a complete

random attack is 20. Similar conclusions can be drawn from other comparisons from Tables 3 and 4. To further compare the effects of attack strength (complete attack *vs* partial attack) and attack intelligence

(targeted attack *vs* random attack), we compare the outcomes of a complete random attack and a partial targeted attack. The results show that the required percolation stages to reach a same degradation level with a complete random attack is more than a partial targeted attack. For instance, it takes 22 and 4 stages for a complete random attack and a partial targeted attack (based on out-degree weight) to reach 80% of degradation in *FR*, respectively. These results reveal that intelligent attacks are more likely to result in severe consequences than powerful attacks, which call for the demanding needs for securing system information. To further illustrate and demonstrate which factor dominates severity of attacks, strength or information, Fig. 9 depicts snapshots of the residual graph at three representative stages with a complete random attack and a partial targeted attack. It can be observed that the partial targeted attack outperforms the complete random attack in rate of node eliminations. This study reinforces the merits of securing system information for robustness enhancement.

4.4. WHFGT based optimal dependency allocation

Apart from identifying influential functionalities and evaluating robustness, the proposed framework can aid in deriving the optimal dependency between processes across infrastructures that would result in enhanced robustness. The optimal dependency (weight) allocation strategy resulting in the optimized robustness metrics can be obtained by solving an optimization problem. For example, the objective could be to maximize the number of stages needed to degrade a selected robustness index by 80% (as a higher number of stages implies higher robustness). Particularly, the nodes (functionalities) that have multiple inputs, i.e., functionalities that depend on various other nodes, are the candidate nodes for this optimization problem with the weights of their incoming edges as the variables that need to be optimized. Further, the sum of the incoming edge weights for any node must be equal to one. For example, We assume that weights can only be selected from the set of {0.0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0}. Enumeration-based method is applied to obtain the optimal solution of this NP-hard problem with the objective of maximizing the required steps to degrade *SR* to a certain level, considering complete targeted attacks as an example. This approach can be extended to other robustness metrics and attack strategies as well. The detailed procedure is illustrated in Fig. 10, where weights of edges are selected randomly from a finite set, while ensuring that the sum of the incoming weights for each candidate node (nodes with multiple dependencies) equates to one. Then, complete targeted attack is simulated to track the number of attacks desired for 80% degradation of the selected robustness index and this process is repeated for multiple times with different edge weight combination. Finally, the weight combination that embodies the highest robustness is the desired solution. In this case, for node 3 that represents the functionality of “Water treatment facility WTP1 is working properly”, the obtained optimal dependency weights are 0.3 and 0.7 for node 35 (“Transit power from GPP to WTP1”) and node 36 (“Transit power from SPP to WTP1”), respectively. Compared to the approach based on random assignments of weights, we can observe a 12% reduction of stages required to degrade *SR* by 80%. This result demonstrates that the proposed approach has great potential to be used as an effective and efficient tool to determine robustness enhancement strategies. The enumeration-based method shows one potential aspect of exploiting WHFGT for robustness enhancement. Exploring other WHFGT based strategies for robustness enhancement is considered as our future research direction.

5. Conclusion

This paper presents, for the first time, a weighted Hetero functional graph theory based framework for modeling and robustness assessment of urban integrated infrastructure networks, including electricity, water, district heating, natural gas, road transportation networks and relevant services. Specifically, system functionalities are denoted by nodes

and edges represent the dependencies among various processes. We conduct various types of attack simulations on the proposed weighted Hetero functional graphs and provide in-depth information about the trajectories of graph fragments. In particular, to simulate real-world scenarios, four attack scenarios, namely complete random attack, complete targeted attack, partial random attack and partial targeted attack, are implemented on the network and the resulting vulnerability is analyzed via plotting the trajectories of several robustness metrics. The outcomes of robustness analysis can be used to determine the optimal dependencies for robustness enhancement. From the results, it can be concluded that the proposed weighted Hetero functional graph based framework and robustness metrics can provide precise insights into the robustness of interdependent infrastructure networks, which highlights the great importance of infrastructure hardening and securing network information. In particular, it finds that partial targeted attack causes greater damage than the complete random attack, which indicates that securing system information is more crucial than physical hardening.

Developing a rigorous and comprehensive weighted Hetero functional graph theory based framework for robustness enhancement will be carried out in the future. Besides, other practical measures for robustness improvement, such as the capability of forming new dependencies after contingencies, will be incorporated into our future research works. Finally, cyber infrastructure for operation and control of integrated urban infrastructure will be further modeled using weighted Hetero functional graph based framework in the future.

CRediT authorship contribution statement

Sai Munikoti: Conceptualization, Methodology, Software, Investigation, Formal analysis, Writing - original draft. **Kexing Lai:** Conceptualization, Methodology, Supervision, Writing - review & editing. **Balasubramaniam Natarajan:** Conceptualization, Formal analysis, Supervision, Writing - review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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